# Two-Stage Neural Passage Ranking:
# An Ensemble Approach for MS MARCO

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## Abstract

We present our solution for the AIML Hackathon 2526 Passage Ranking Challenge, achieving a MAP@10 of **0.7233** on the public leaderboard. Our approach follows a two-stage retrieval pipeline: (1) a Dual Encoder using `all-mpnet-base-v2` retrieves 300 candidate passages per query via FAISS, and (2) a Cross-Encoder ensemble of three state-of-the-art rerankers (BGE-M3, Jina, MiniLM) refines the ranking using Reciprocal Rank Fusion (RRF). Through systematic ablation studies and grid search optimization, we found that an unweighted ensemble with minimal smoothing ($k$=1) generalizes better than weighted configurations, despite achieving slightly lower validation scores.


## 1. Introduction

Passage ranking is a core task in information retrieval, where the goal is to rank relevant passages higher for a given query. The AIML Hackathon challenge uses a sampled subset of MS MARCO, requiring participants to rank passages and submit the top-10 results per query, evaluated via Mean Average Precision at 10 (MAP@10).

Modern neural retrieval systems typically employ a two-stage architecture: a fast first-stage retriever generates candidate passages, followed by a more expensive reranker for precision refinement. This design balances efficiency and effectiveness.

Our contributions include: (1) a systematic analysis of the retrieval cutoff parameter $K$ using Oracle Recall; (2) an ablation study comparing four cross-encoder rerankers; (3) a grid search over ensemble configurations revealing that simpler, unweighted fusion outperforms complex weighted schemes on unseen data.

## 2. Method

### 2.1. First-Stage Retrieval: Dual Encoder

We use a Bi-Encoder (Dual Encoder) architecture where queries and passages are encoded independently into dense vectors. Given query $q$ and passage $p$:

$$\text{score}(q, p) = \mathbf{e}_q^\top \mathbf{e}_p \tag{1}$$

where $\mathbf{e}_q, \mathbf{e}_p \in \mathbb{R}^d$ are the L2-normalized embeddings.

**Model:** `sentence-transformers/all-mpnet-base-v2` (768-dim embeddings). We encode the full passage collection once and build a FAISS Flat Inner Product index for efficient retrieval.

**Cutoff Selection:** We analyzed *Oracle Recall*: the fraction of queries for which the relevant passage appears in the top-$K$ candidates. At $K$=300, Oracle Recall reached $\sim$0.99, meaning nearly all relevant documents are retrieved. Increasing $K$ beyond 300 added computational cost without improving coverage.

### 2.2. Second-Stage Reranking: Cross-Encoders

Cross-Encoders jointly encode the query-passage pair, enabling richer interaction modeling:

$$\text{score}(q, p) = f_\theta([\texttt{CLS}]\ q\ [\texttt{SEP}]\ p\ [\texttt{SEP}]) \tag{2}$$

We evaluated four models on the 300 candidates per query:

- `jinaai/jina-reranker-v2-base-multilingual`
- `BAAI/bge-reranker-v2-m3`
- `cross-encoder/ms-marco-MiniLM-L-12-v2`
- `castorini/monot5-base-msmarco-10k`

### 2.3. Ensemble: Reciprocal Rank Fusion

To combine rankings from multiple rerankers, we use Reciprocal Rank Fusion (RRF):

$$\text{RRF}(d) = \sum_{i=1}^{N} w_i \cdot \frac{1}{k + r_i(d)} \tag{3}$$

where $r_i(d)$ is the rank of document $d$ in model $i$, $k$ is a smoothing constant, and $w_i$ are optional weights.

We explored two fusion strategies:

- **Unweighted:** $w_i = 1$ for all models
- **Weighted:** $w_i \propto \text{MAP}_i$ (proportional to individual model performance)

# 3. Experiments

## 3.1. Ablation Study: Individual Rerankers

Table 1 shows the validation MAP@10 for each cross-encoder applied independently to the 300 candidates from MPNet.

| Model | MAP@10 |
|---|---|
| Jina Reranker v2 | 0.696 |
| BGE Reranker v2-M3 | 0.690 |
| MiniLM-L12 | 0.680 |
| MonoT5 | 0.626 |

*Table 1.* Individual reranker performance on validation set.

MonoT5 underperformed significantly and was excluded from the final ensemble. We also tested BM25 as a lexical baseline, but its MAP ($\sim$0.45) was too low to contribute meaningfully.

## 3.2. Grid Search: Ensemble Configurations

We conducted a grid search over: (1) model subsets, (2) weighted vs. unweighted fusion, (3) smoothing parameter $k \in \{0, 1, 2, 5, 10, ..., 100\}$, and (4) inclusion of BM25.

**Key finding:** The best local validation score (0.7052) came from a *weighted* ensemble including BM25. However, this configuration achieved only 0.7219 on the public leaderboard, showing slightly worse generalization performance.

## 3.3. Final Configuration

The winning configuration was simpler:

- **Retrieval:** MPNet, $K$=300
- **Rerankers:** Jina + BGE + MiniLM (Top-3)
- **Fusion:** Unweighted RRF, $k$=1

| Config | Models | Val | LB |
|---|---|---|---|
| Weighted+BM25 | 4 | 0.7052 | 0.7219 |
| **Unweighted Top-3** | **3** | **0.7032** | **0.7233** |

*Table 2.* Comparison of ensemble configurations. Val = local validation MAP@10; LB = public leaderboard.

The unweighted configuration achieved a lower validation score but generalized better to unseen data, achieving our best leaderboard result of **0.7233**.

# 4. Discussion

**Why simpler is better.** Adding BM25 and using MAP-based weights optimized for the validation set but reduced generalization. The top-performing neural models (Jina, BGE, MiniLM) are all strong and complementary; giving them equal votes via unweighted RRF prevented any single model from dominating and introduced beneficial diversity.

**Smoothing parameter.** The optimal $k$=1 emphasizes top ranks heavily. With $k$=1, the first-ranked document receives score $1/2$, while the 10th receives $1/11$: a 5.5× difference. Higher $k$ values flatten this distribution, reducing discrimination.

**LLM-based reranking attempts.** We experimented with using instruction-tuned language models (QWEN 3B fp16, QWEN 7B/14B int4-quantized) as a post-processing step after cross-encoder reranking. However, these approaches degraded performance compared to the pure cross-encoder ensemble, likely due to the models' general-purpose nature versus the task-specific training of dedicated rerankers.

**Limitations.** Our pipeline assumes a fixed collection. We did not explore query expansion, passage chunking, or fine-tuning the retriever on in-domain data, which could further improve results.

# 5. Conclusion

We developed a two-stage passage ranking system combining dense retrieval with an ensemble of cross-encoder rerankers. Our systematic experiments revealed that:

1. $K$=300 candidates provide near-perfect Oracle Recall with MPNet
2. The top three rerankers outperform four including a weaker model
3. Unweighted RRF generalizes better than weighted schemes

The final system achieved MAP@10 of 0.7233 on the public leaderboard, demonstrating that careful model selection and ensemble design can outperform more complex configurations.